Automation & Upgrade of LoadX

End to End Business Analyst Case Study in Operations

Safar Logistics & Warehousing | Brownfield Project

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EXECUTIVE SUMMARY

Safar logistics processes 90,000 -150,000 truck dispatches daily. A critical safety checkpoint before it dispatches is verifying load security and it was digitized in 2018 via an internal tool called LoadX, the tool still depends heavily on manual effort - images are opened via links, job allocation is done by managers, KPIs are calculated in excel and audit feedback takes days to reach associates.

This case study defines the full business analysis for upgrading LoadX into a structured, automated, and data - driven system. The solution eliminates manual intervention across job assignment, quality review, audit sampling, KPI reporting and feedback loops, enabling Safar to scale its daily review capacity, improve accuracy and achieve near real time operational visibility.

Problem in Brief

- Associates process only 300 images/day against a daily volume of 100,000+ due to manual workflow
- Ops & Audit managers spend significant amount of time consolidating logs and computing KPIs, delaying reporting
- Audit team reviews only 5-10% of jobs. Feedback reaches associates days later, allowing errors to repeat
- No centralized data store - job, audit and performance records are scattered across spreadsheet

Proposed Solution

- Auto-assignment of jobs to associates and audit associates - no manual allocation
- Inline structured decision options and image display - no link opening required
- Live SLA timer with auto-expiry and reassignment at 2 - minute threshold
- Automated defect emails triggered instantly on audit-associate decision mismatch
- Role based real-time dashboard for all user levels from associate to VP operations
- Centralized data store for all job, audit and performance records with export capability

Projected Impact

3x increase	97%	30 secs	70%	10 - 15%
Associate Throughput	Projected Accuracy	Average TAT	Error Rate Reduction	Audit Coverage

The upgrade is projected to increase associate throughput 3x - from 300 to 840 jobs per associate per day. While improving accuracy from 90% to 97% through real-time feedback, structured decision workflows and automated audit coverage

Phased Delivery Plan

Sprint	Deliverable
Sprint 1	Core review screen, auto-assignment, SLA timer
Sprint 2	Audit workflow, defect email automation
Sprint 3	Role-based dashboards, access control, export
Sprint 4	UAT & go live

Key Risks & Gaps Flagged

- SOP standardization and decision option ambiguity - known pain point, deferred to phase 2
- Audit escalation path beyond Audit Manager not defined - phase 2 item
- Dedicated post go-live support window recommended to capture associate reported issues

1. Organization & Business Problem Overview

1.1 Organization Overview

Safar is a third - party Logistics and Warehousing company operating globally. It handles entire supply chain including storing, inventory, picking and packing items and providing transportation for last mile - delivery. On a daily basis, company dispatches around 90,000 to 150,000 loaded trucks for delivery.

1.2 Business Overview & Problem Context

A critical SOP before dispatch requires verifying that every truck is properly secured.Initially this was done manually by a Transportation manager. In, 2018, Safar introduced LoadX to digitize the process. However, the current implementation still relies heavily on manual efforts across all roles

The current (As-Is) process flow:

1. **Image capture and queue:** Truck associate captures and uploads the load image. System validates image quality and queues it for review
2. **Manual review:** Ops manager manually assigns images in batches to LoadX associates. Each associate opens the job through a link, manually record their decision and sends logs back to the Ops Manager
3. **Manual Reporting & Audit Sampling:** Ops Manager consolidates logs, manually calculates TAT and accuracy and sends a sample to the Audit Manager. The audit manager similarly assigns sample jobs manually to audit associates and computes quality metrics.
4. **Delayed feedback loop:** Audit associates identify errors and report to Audit Manager, who relays findings to the Ops Manager, who then passes them to LoadX associates, by the time which errors may already have repeated

1.2.1 Business Impact

- Associates generate reduced output of 300 jobs/person/day due to high manual efforts
- Managers spend significant time consolidating logs and computing KPIs, delaying reports & performance management
- Inconsistent quality due to delayed feedback as mistakes may already have repeated by the time it reaches associates
- Overall - high operational costs, slower processing, delayed reporting and limited scalability

1.2.2 Problem Statement

- With 100,000+ daily images, associates can process only around 300 images per day within 7 productive hours, resulting in high TAT.
- Accuracy levels remain around 90% due to delayed feedback and inconsistent SOPs interpretation
- Managers spend time on calculating KPIs & preparing reports, instead of performance management
- Audit team reviews only 5–10% of samples due to volume constraints, leading to limited quality control.
- The existing LoadX system does not act as a centralized system of record to review data.
- Overall, the process is time-consuming, resource-intensive, and not scalable

2. Stakeholder Analysis

Stakeholders were identified by reviewing the process and listing all users involved in the workflow, as well as teams supporting the system.

Stakeholder Register Table

Stakeholder	Group	Interest/Impact	Influence	Priority
VP Operations	Primary	Approves budget & scope. Monitors ROI	High	High
Ops Manager	Primary	Benefits from auto-reporting	High	High
LoadX Associate	Primary	Core end user of upgraded review screen	Medium	High
Audit Associate	Primary	Receives auto assigned audit jobs, triggers defect mails	Medium	High
Audit Manager	Primary	Configures audit sample %, oversees quality trends	High	High
Product Manager	Secondary	Co-signs requirements, manages sprint	High	High
Development Team	Secondary	Builds and delivers the upgraded system	Medium	Medium
QA Team	Secondary	Designs and executes UAT scenarios	Medium	Medium
Training Team	Secondary	Trains users, creates updated SOP	Low	Medium
IT/Infrastructure Team	Secondary	Manages browser access	Low	Low
Data Analytics Team	Secondary	Own data pipeline to LoadX backend	Medium	Medium

3. Business Requirements (BRD)

3.1 Business Objective

- To reduce the manual effort and increase associate throughput
- To provide associates and managers with real time visibility into KPIs
- To provide immediate and real time feedback to associates on the processed jobs
- To improve overall process efficiency by optimizing resources and scalability
- To have a single centralized system of record for all the process data

3.2 Business Rules

- Every truck image must be reviewed and approved by LoadX associate before dispatch
- A defect is raised when decision of LoadX associate's decision differs from Audit associate's decision
- Jobs must be completed within 2 mins as per SLA
- Submitted job decisions are final
- Audit sampling percentage and accuracy thresholds must be defined and controlled
- Poor quality images must not be processed

3.3 Scope & Assumptions

3.3.1 In scope

- Auto assignment of jobs to associates, replacing manual distribution.
- Auto-display of truck images sourced from data pipeline, no separate link required.
- Inline structured decision options and image quality flagging
- Session based workflows for associates with post session performance summary
- Live SLA timer for each job with auto-expiry and reassignment on breach
- Automated defect email on audit associate decision mismatch
- Role-based dashboards with real-time KPIs
- Exportable reports in Excel & Pdf.

3.3.2 Out of scope

- Resolution of SOP ambiguity and SOP standardization
- Audit escalation path beyond Audit Manager
- Mobile Application support
- Historical data migration from previous system.
- SSO or authentication system changes (existing login system assumed).
- Payroll, HR, or performance appraisal integration, data is used only for ops improvement
- Real-time SLA breach notifications to Ops Managers beyond dashboard visibility.

3.3.3 Assumptions

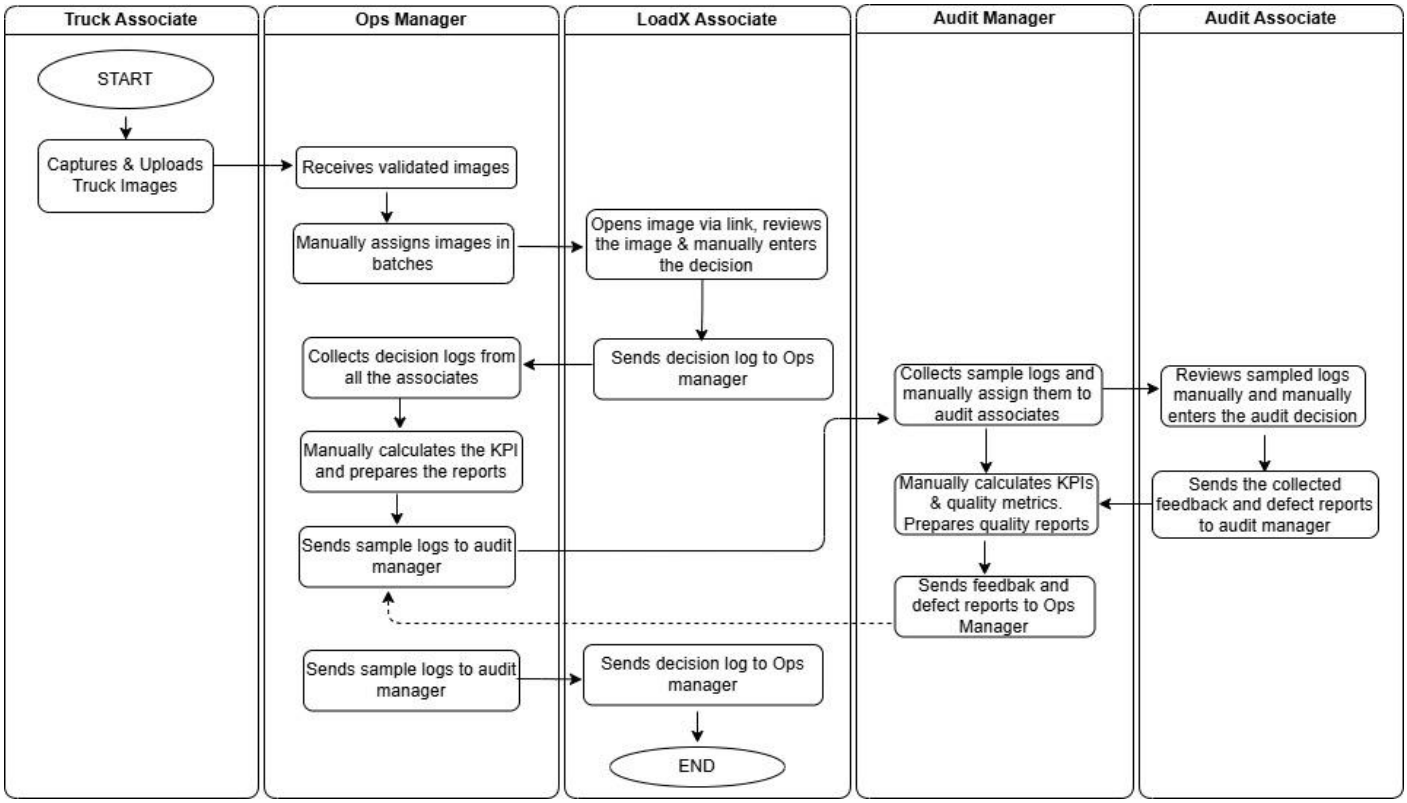
- All users have access to work laptops with Chrome or Firefox installed.
- Audit Manager is a distinct role from Ops Manager with separate system access.
- Existing user authentication infrastructure will be reused — no new login system required.
- UAT will be led by the QA Team with support from the Ops Manager for associate testers.
- The Data Analytics team will own and maintain the backend pipeline where LoadX data flows

3.4 Business Requirements

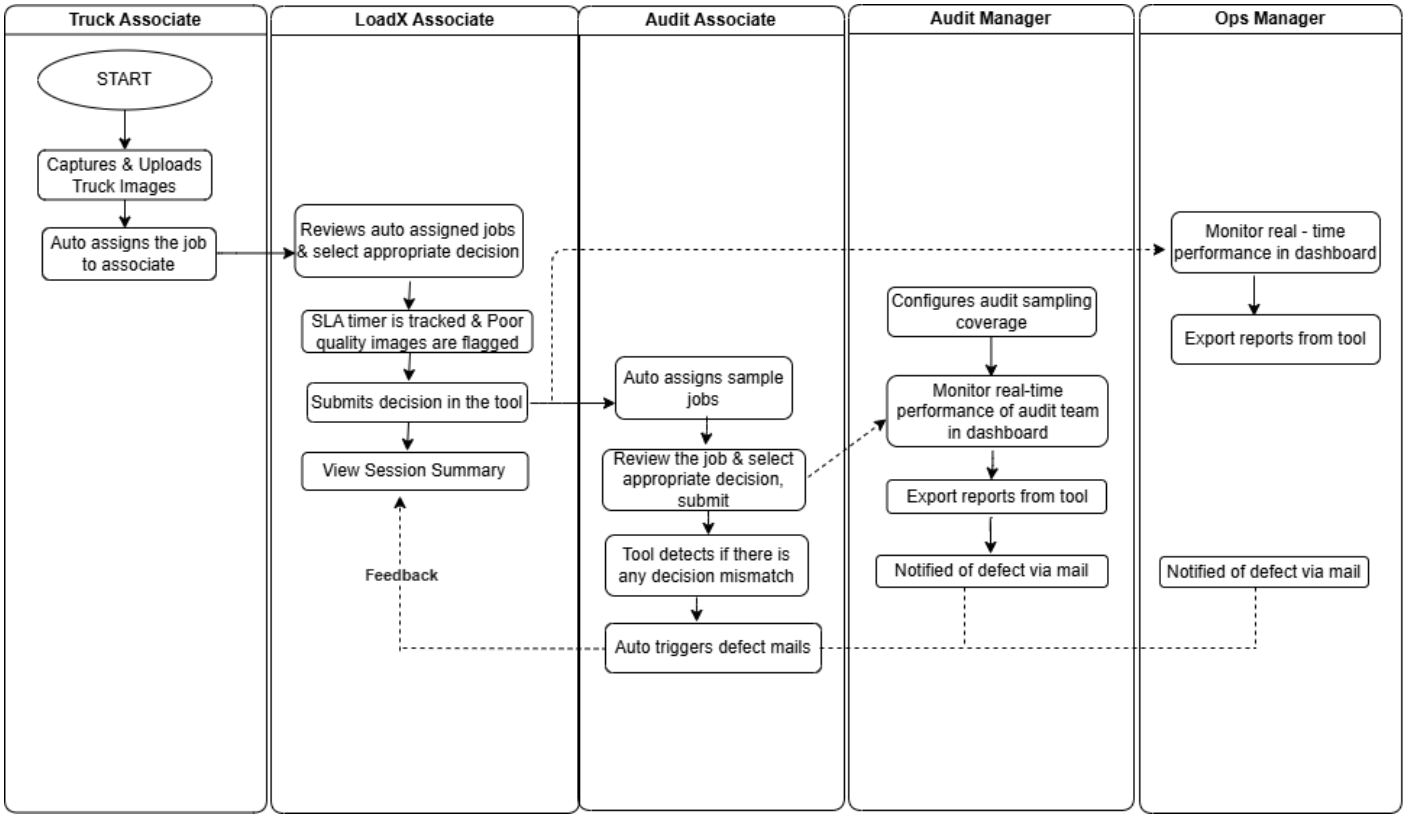
BR - ID	Requirement	Priority	Raised by
BR-01	Eliminate manual effort in job allocation, job viewing and tracking across the LoadX process	High	Ops Manager
BR-02	Enable faster processing of image review jobs to improve overall throughput	High	VP operations
BR-03	Provide real time visibility into associate and process metrics	High	Ops Manager
BR-04	Ensure immediate feedback loop for associates to improve KPI and reduce errors	High	Audit Manager
BR-05	Establish a centralized system of record for all job, audit and performance data	High	Data Analytics
BR-06	Improve overall process accuracy through structured audit and validation mechanisms	High	Audit Manager
BR-07	Reduce TAT for job completion to meet SLA requirements	High	Ops Manager
BR-08	Ensuring system can support increasing daily job volumes without impacting process efficiency	Medium	VP Operations
BR-09	Reduce dependency on manual reporting and external tools for KPI tracking	Medium	Ops Manager
BR-10	Increase audit coverage and effectiveness without significantly increasing manual effort	High	Audit Manager

3.5 Process Flows

3.5.1 As - Is (Current State)



1.5.2 To - Be



4.User Stories/Acceptance Criteria (FRD)

4.1 Functional Requirements

FR - ID	Requirement	Priority	Raised by
FR-01	System shall auto assign jobs from the queue to associates that are logged in tool during active shift hours. Jobs will be distributed evenly among associates based on availability to ensure balanced workload	High	Ops Manager
FR-02	Initial implementation will follow random sampling of 10-15% and sampling percentage is configurable. Advanced sampling strategies may be introduced in future phases.	High	Audit Manager
FR-03	System shall auto display the image of truck on the screen	High	Associate
FR-04	System shall allow structured decision options for associates	High	Associate
FR-05	System shall allow associates to flag poor quality images and remove them from processing queue	High	Associate
FR-06	System shall display live timer in the tool for each job	High	Ops Manager
FR-07	System shall auto expire the jobs exceeding SLA time threshold. Expired jobs will be re-queued for reassignment.	High	Ops Manager
FR-08	System shall provide session - based workflows for associates, enabling start and end session tracking	Medium	Product Manager
FR-09	System shall generate session summary after ending the session	High	Ops Manager
FR-10	System shall trigger automated defect mails when LoadX associate and audit associate decision mismatch	High	Audit Manager
FR-11	System shall provide role-based dashboard with real time KPIs	High	VP Ops
FR-12	System shall allow export of reports in Excel/PDF format	Medium	Ops Manager
FR-13	System shall maintain centralized database storing all job, audit, image and performance data	High	Data Analytics

4.2 Non-Functional Requirements

NFR - ID	Category	Requirement
NFR-01	Performance	Truck image shall load within 2 seconds of job assignment
NFR-02	Scalability	System shall support processing 200,000 images per day
NFR-03	Portability	System shall be accessible on chrome and firefox on desktop
NFR-04	Security	Users shall access only data and features permitted by their role
NFR-05	Availability	System shall have 99.5% uptime excluding planned maintenance
NFR-06	Reliability	System shall ensure no data loss during job processing
NFR-07	Data Integrity	All job decisions must be stored without duplication or overwrite

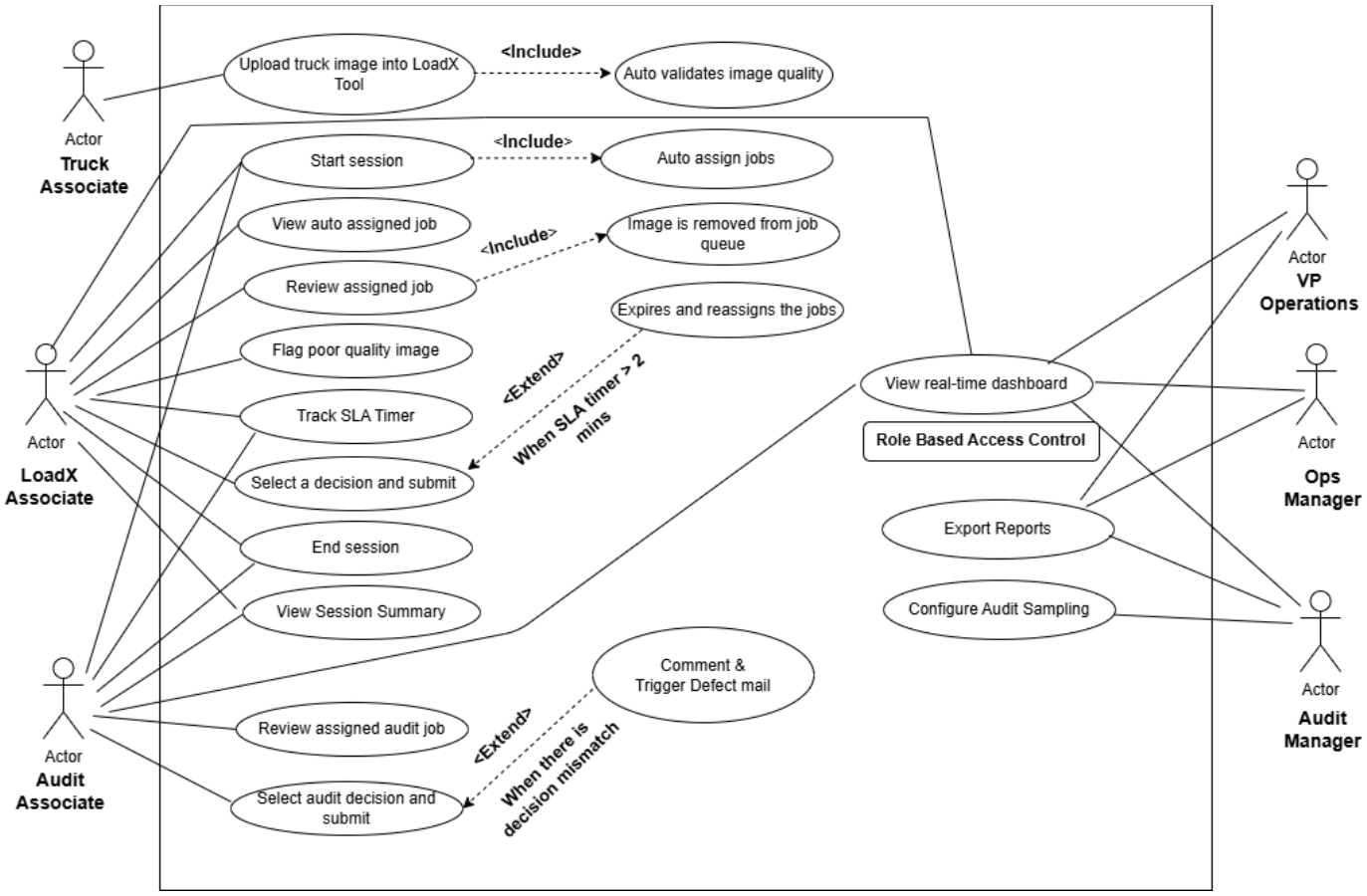
4.3 User Stories and Acceptance Criteria

Full 16 user stories with acceptance criteria are available in Appendix A.

US02	**As a LoadX associate, I want structured decision options, so that I can reduce errors and ensure consistent decision making**		
	Acceptance Criteria		
	Scenario 1 Given decision options displayed When associate selects one option Then the decision should be recorded successfully upon submission	**Scenario 2** Given decision options displayed When associate don't select any option and submits the job Then the system should show a validation error	**Scenario 3** Given decision options are displayed When associates try to select more than one option Then the system should restrict selection to one option
US11	**As an Audit associate, I want defect mails to be automatically triggered when decision mismatch, so that LoadX associates get real time feedback and errors are tracked**		
	Acceptance Criteria		
	Scenario 1 Given job is completed and audit decision differs from LoadX associate decision When the audit is submitted Then a defect should be automatically created and notification mail should be triggered to associates & manager	**Scenario 2** Given a defect is triggered When notification email is sent Then the email should include - Job ID, Job Link, Associate ID, Associate decision, Audit Associate ID, Audit Decision and Comment	**Scenario 3** Given a defect is triggered When notification email fails to send Then system should log the failure and allow retry and tracking of unsent notifications
US14	**As an Ops Manager, I want all job, audit and performance data to be stored in a centralized system, so that I can access accurate and consistent data for reporting and analysis**		
	Acceptance Criteria		
	Scenario 1 Given jobs are processed by associates and audits are completed When the data is generated Then all job, audit and performance data should be stored in a centralized system	**Scenario 2** Given data is stored in a centralized system When Ops Manager access reports or dashboards Then data should be available and reflect real - time data & have an option to export the data in excel & pdf	

4.4 Use Case Flow

LoadX Tool Use Case

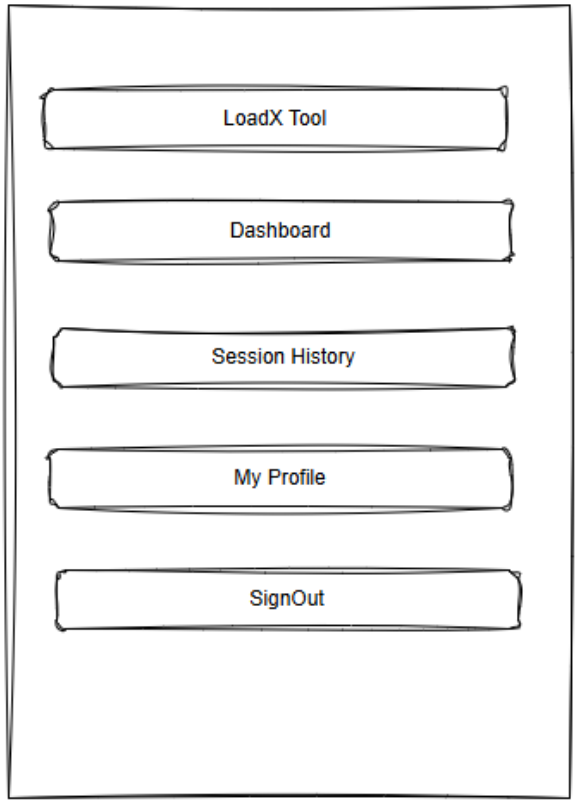


5. UI/UX Requirements and Screen Layout

5.1 UI/UX objectives

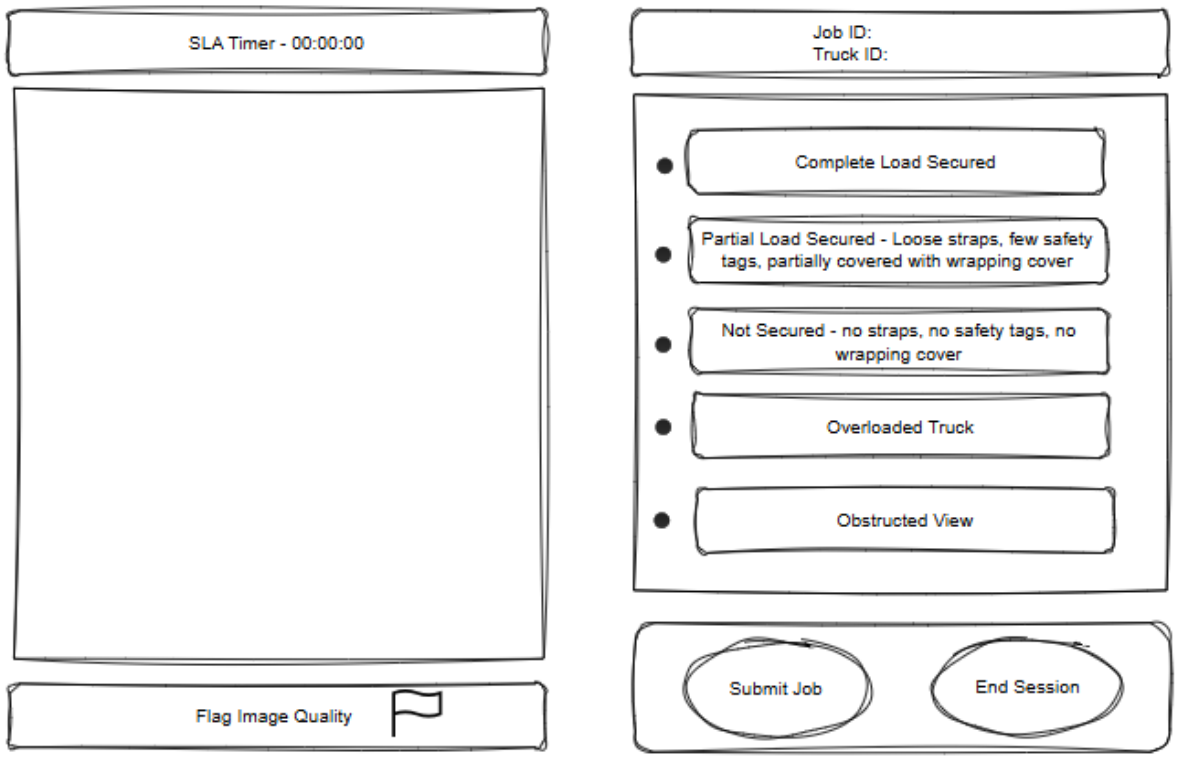
- Minimizing cognitive load - associates process high volume of jobs, interface should be fast, clear and require minimal navigation
- Structured inputs only - no - free text decision fields to prevent inconsistent data entry
- Immediate Feedback - SLA timer, session metrics and submission confirmations must be real - time
- Role based views - each role sees only what is relevant to their functions

5.2 LoadX Tool Page Screen



Welcome to LoadX

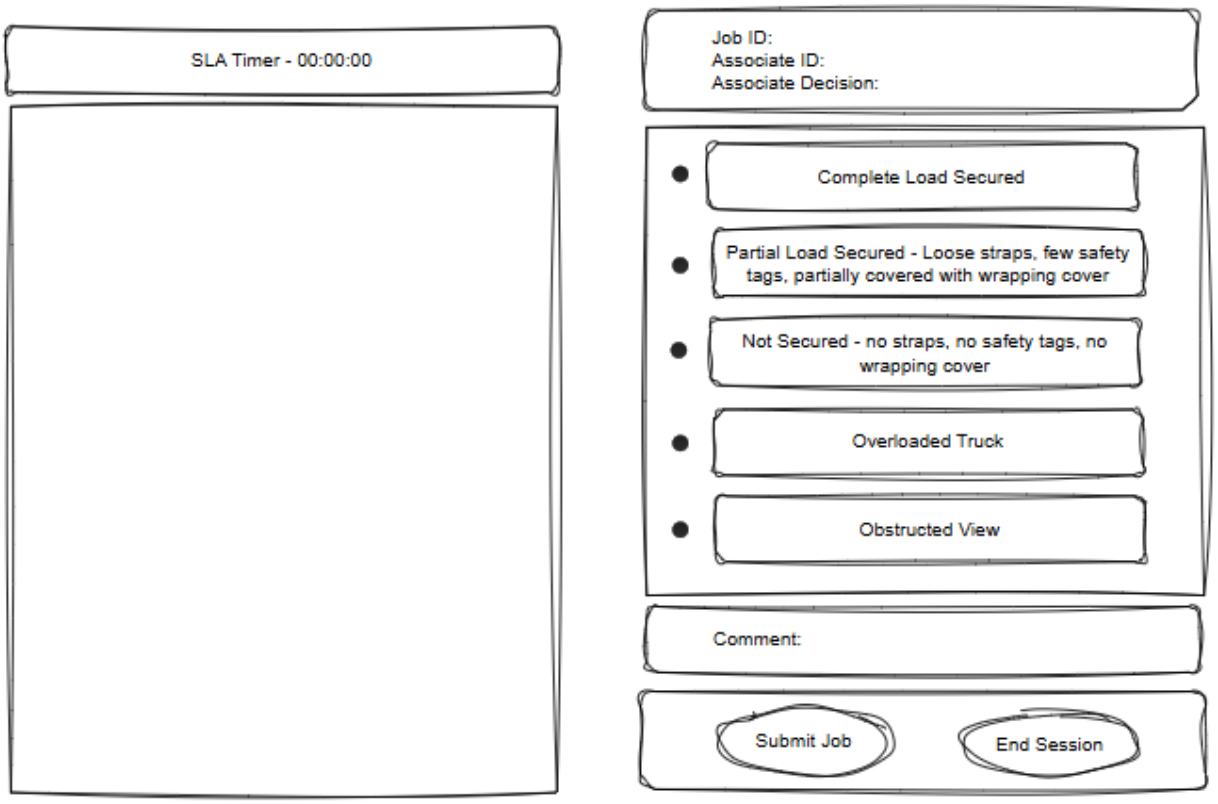
5.3 LoadX Associate Screen Layout & Specifications



Specifications

- Truck image is displayed full-width within the panel
- Image renders automatically, no link to open
- Flagging opens a confirmation modal
- Image cannot be zoomed or manipulated as zooming might lead to inconsistent decisions & also increase TAT due to micro analysis
- Submit button enabled only after decision is selected or image is flagged for quality
- After submission, next job loads automatically

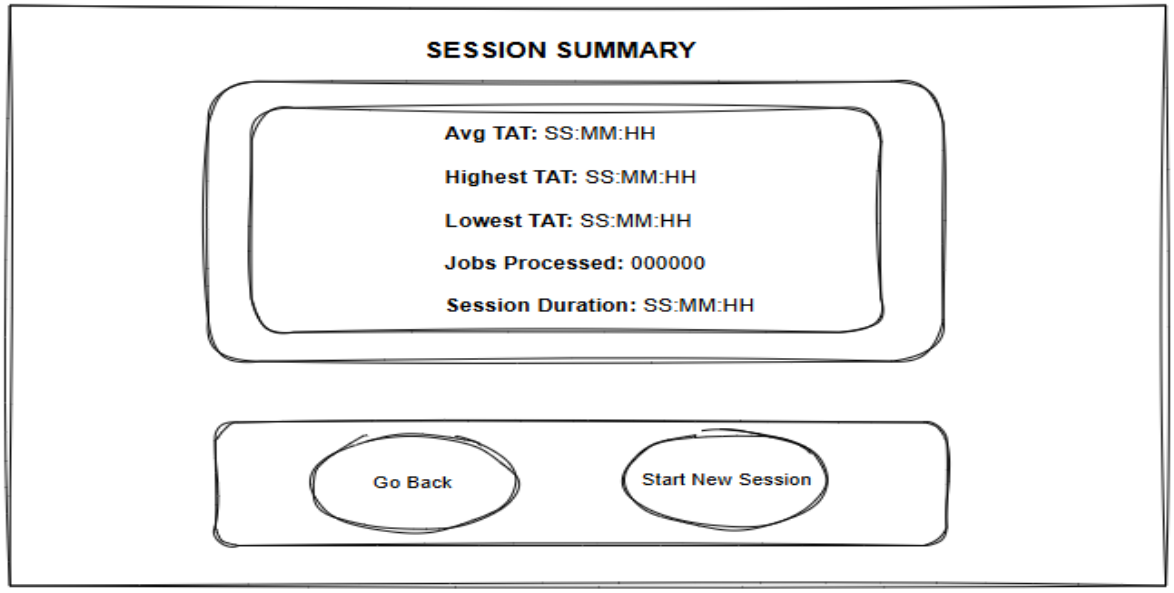
5.4 Audit Associate Screen Layout & Specifications



Specifications

- Truck image is displayed full-width within the panel
- Image renders automatically, no link to open
- Flagging opens a confirmation modal
- Image cannot be zoomed or manipulated
- Submit button enabled only after decision is selected or image is flagged for quality
- After submission, next job loads automatically

5.5 Session Summary & Specifications



Specifications

- Displayed automatically when associate or audit associate ends a session
- Session history accessible from the associate's profile
- If no jobs were processed in a session, in the place of summary "No activity in the session" should be displayed
- Go back should lead to main page of tool

6. UAT & Testing

6.1 UAT Approach

UAT will be led by the QA team, who will design test case scenarios and cases based on Acceptance criteria defined in user stories. Ops Manager will be responsible for coordinating associate participation as business testers. UAT will be conducted in staging environment that mirrors production data and access controls

6.1.1 UAT Entry Criteria

- All user stories for the sprint have passed QA's internal testing
- Staging environment is set up with test accounts for all roles
- Test data is loaded into staging environment (sample images & job queues)
- UAT test cases should be signed off by Ops and Audit managers

6.1.2 UAT Exit Criteria

- All critical and high priority test cases pass with no open defects
- Medium priority defects are documented with agreed resolution timelines
- Sign-off received from VP operations and Product Manager

6.2 UAT Test Cases

There are 13 high level scenario test cases and all of them are available in [Appendix B]

TC - ID	US - ID (Reference)	High Level Scenario	Expected Outcome	Tester Role	Priority
TC-01	US-01	Associate opens an assigned job & views the screen	Image loads, with inline decision options	LoadX Associate	Critical
TC-04	US-04	Associate flags an image as poor quality	Recorded image quality and removed from queue	LoadX Associate	Critical
TC-09	US-10	Audit associate submit audit decision for a job where there is no decision mismatch	Audit decision is recorded successfully	Audit Associate	Critical
TC-11	US-13	Ops Manager views dashboard for performance monitoring	Display of real-time KPIs	Ops Manager	Critical
TC-12	US-14	System stores and retrieves operational data	All job, audit and performance data are stored centrally and can be accessed	Ops Managet	Critical
TC-13	US-15	Audit Manager configures audit sampling percentage with valid percentage	Percentage is saved and applied to future jobs	Audit Manager	High

7. Data & Reporting

Data Analytics team owns data pipeline, validates data accuracy, KPI definitions and any other inconsistencies and hygiene of data. A full data dictionary covering all 30+ data points including field definitions, data types, and sources are available in Appendix C

7.1 KPI Definitions

KPI	Formula/Logic	Data required
Accuracy	(No. of jobs completed - no. of defects) * 100/Total number of jobs completed	Jobs_completed_count, defect_count
Avg TAT	SUM (Decision_Timestamp - Assignment_Timestamp)/No. of jobs completed	Decision_Timestamp, Assignment_Timestamp, Jobs_completed_count
Highest TAT	Maximum of (Decision_Timestamp - Assignment_Timestamp) in a session	Decision_Timestamp, Assignment_Timestamp
Lowest TAT	Minimum of (Decision_Timestamp - Assignment_Timestamp) in a session	Decision_Timestamp, Assignment_Timestamp
Session Duration	Session_End_Time - Session_Start_Time	Session_End_Time, Session_start_time
SLA Breach %	(Count of Expired Jobs/Total jobs completed) *100	Is_Expired, Jobs_completed_count
Productivity rate	Jobs completed/Session Duration	Jobs_completed_count, Session_Duration
Productive Hours	Sum (session durations) of an associate in a day	Session _Duration
Defect rate	(Count of (Is_mismatch = Yes)/Total Audited jobs)*100	Is_mismatch, Audit_ID
Audit coverage	(Audited jobs/Total_jobs_completed)*100	Audit_id, job_id
Session count	Count of (Session_ID) in a day	Session_ID
Expired Job count	Count of (Is_Expired = Yes)	Is_Expired
Flagged Job count	Count of (Is_Flagged= Yes)	Is_Flagged
Reassignment rate	Total reassignments/Total_jobs_completed	Reassignment_count, Total_jobs_completed

7.2 Role Based Dashboard KPIs

User Role	KPI & Metrics Visible
LoadX Associate	Jobs processed, Avg TAT, Productivity Hours, Accuracy Rate, Defect count, Expired Jobs, Number of sessions, Flagged jobs count
Audit Associate	Jobs audited, Avg TAT, Productivity Hours, Audit accuracy rate, Defects Flagged, Number of sessions, Defect category
Ops Manager	Team-level and Individual level total jobs processed, team avg TAT, Productivity Hours, Team accuracy rate, Team defect count, flagged image count, SLA breach count, Flagged jobs count, Defect category. Jobs in Queue. With shift and region fliter

Audit Manager	Audit coverage % (configured vs actual), defect rate trend, top defect categories, associate-level defect frequency, audit associate performance, configurable audit % setting.
VP operations	All Ops Manager KPIs, cross-region/shift comparison, daily/weekly/monthly trend views, overall system throughput vs volume, cost efficiency metrics.

7.3 Reporting Requirements

- All reports must be exportable to excel(.xlsx) and PDF format
- Export option is available to Ops and Audit managers, VP operations
- Dashboard data must refresh in real time, no manual refresh is required
- Dashboards must include filters for date range, shift, and region (for ops and above)
- Historical report data must be retained in the system for a minimum of 12 months

8. Recommendations

8.1 Projected Business Impact

The following projections are based on process analysis conducted as part of this case-study. Baseline metrics are derived from real-world logistic operations. Projected outcomes are estimated based on industry benchmarks and process improvement principles.

Metric	As - Is	To-Be	Outcome
Associate Throughput	300 jobs/associate/day	840jobs/associate/day	3x increase
Feedback Loop	1-2 Days (manual relay)	Under 5 minutes (automated defect mail)	98% faster
Audit Coverage	5-10% (manual, inconsistent)	10-15% (automated & configurable)	Minimum 10% of jobs are audited
Accuracy	Around 90% (delayed feedback)	97% (real-time feedback)	7pp improvement. 70% error rate reduction
Average TAT	Around 84 secs, untracked, no SLA	Around 30 secs, 2 min SLA with auto-expiry	64% reduction in TAT
Real Time Reporting	Manual report consolidation	Real-time dashboards + auto generated reports	Zero manual reporting effort

8.2 Recommendations & Phase 2 Considerations

- Adopt a phased sprint approach:
  Sprint 1 - Core review screen + auto-assignment + SLA timer
  Sprint 2 - Audit workflow + Defect Email
  Sprint 3 - Dashboards, Role based access control, export
  Sprint 4 - UAT & Go live production
- Conduct a training session for associates and managers before going live by training team to ensure tool training and any updated SOP
- Define and agree on structured decision options with Ops Manager and VP operations before development starts
- Flag audit escalation path, audit selection method and SOP standardization as Phase 2 enhancement in product roadmap as these are the gaps people will escalate post going live
- Dedicated support should be provided after going live to collect and review associate reported issues.

8.2.1 Phase 2 Considerations

- Audit escalation path, no workflow exists beyond Audit Manager if their decision is also disputed.
- Selection method used for auditing jobs is not very efficient. Alternative methods can be explored
- SOP standardization, decision option ambiguity is a known pain point but out of scope for this phase.

Appendix- A

User Stories and Acceptance Criteria

LoadX User Stories

US01	**As a LoadX associate, I want the truck image to be displayed automatically, so that I can save time and process more jobs efficiently**		
	Acceptance Criteria		
	Scenario 1 Given a job is assigned to the associate When the job is loaded successfully Then the truck image should be displayed automatically	**Scenario 2** Given jobs are available and a job is assigned to the associate When the job fails to load Then an error message should be displayed, associate should flag this and submit	
US02	**As a LoadX associate, I want structured decision options, so that I can reduce errors and ensure consistent decision making**		
	Acceptance Criteria		
	Scenario 1 Given decision options displayed When associate selects one option Then the decision should be recorded successfully upon submission	**Scenario 2** Given decision options displayed When associate don't select any option and submits the job Then the system should show a validation error	**Scenario 3** Given decision options are displayed When associates try to select more than one option Then the system should restrict selection to one option
US03	**As a LoadX associate, I want the next job to automatically assigned and displayed after submission, so that workflow continuity is not interrupted**		
	Acceptance Criteria		
	Scenario 1 Given a job is completed and submitted When additional jobs are available in the queue Then the next job should be automatically assigned without requiring manual refresh or action	**Scenario 2** Given a job is submitted When no jobs are available in the queue Then system should display "No Jobs Available"	

US04	**As a LoadX associate, I want to flag poor quality images, so that invalid jobs are removed from processing queue**	
	Acceptance Criteria	
	Scenario 1 Given a job is assigned and an image is displayed When the associates select “Flag Image” Then the job should be recorded, marked as poor quality and removed from processing queue	**Scenario 2** Given a job is assigned and image is displayed When the associate proceeds without selecting “Flag Image” Then the job should continue for normal decision processing
US05	**As a LoadX associate, I want to see a live SLA timer, so that I can complete jobs within the defined time limit**	
	Acceptance Criteria	
	Scenario 1 Given a job is assigned When the job is loaded Then a timer should start from 0 seconds and continue increasing as time progresses	**Scenario 2** Given a job is in progress When timer exceeds SLA 2 mins threshold Then “Job Expired” message is displayed
US06	**As a LoadX associate, I want jobs to expire if exceeded SLA, so that it is reassigned and process flow is not blocked**	
	Acceptance Criteria	
	Scenario 1 Given a job is in progress When the elapsed time is less than 2 minutes Then the job should remain active and associate should be allowed to select a decision and submit	**Scenario 2** Given a job is in progress When the elapsed time exceeds 2 minutes Then the job should be automatically expired, removed from queue and reassigned to another associate

US07	As a LoadX associate, I want to start and end a session, so that my productivity and work duration are tracked		
	Acceptance Criteria		
	Scenario 1 Given the associate is logged into tool When the associate clicks “Start Session” Then a new session should be initiated and session start time should be recorded	**Scenario 2** Given a session is active When the associate clicks “End Session” Then the session should be closed and session end time should be recorded	**Scenario 3** Given a session has started and ended When the session is completed Then system should calculate total session duration

US08	As a LoadX associate, I want to view my session performance summary after ending the session, so that I can track and improve my productivity and accuracy	
	Acceptance Criteria	
	Scenario 1 Given an active session is in progress When associate ends the session Then session should be ended and session performance summary (Avg TAT, Highest TAT, Lowest TAT, Number of Jobs Processed and Session duration)	**Scenario 2** Given the associate ends a session without processing any jobs When the summary is displayed Then the system should display “No activity in session” message

Audit Associate User Stories

US09	As an Audit associate, I want to view associate id and decision along with image, so that I can validate their work efficiently		
	Acceptance Criteria		
	Scenario 1 Given an audit job is assigned When Audit associate views the job Then the truck image should be displayed, associate id and associate decision should be visible	**Scenario 2** Given an audit job is assigned When associate id or decision is not available Then system should display appropriate message and allow associate to flag the issue	**Scenario 3** Given the associate decision is displayed When the Audi Associate views the job Then the decision shown should match original associate submission

US10	As an Audit associate, I want to record my audit decision and comment, so that discrepancies are properly documented and communicated	
	Acceptance Criteria	
	Scenario 1 Given job is assigned and displayed to audit associate When associate selects audit decision Then system should record audit decision successfully	**Scenario 2** Given audit decision differs from LoadX associate decision When the Audit Associate submits without entering a comment Then system should prompt for a mandatory comment and prevent submission until comment is provided

US11	As an Audit associate, I want defect mails to be automatically triggered when decision mismatch, so that LoadX associates get real time feedback and errors are tracked		
	Acceptance Criteria		
	Scenario 1 Given job is completed and audit decision differs from LoadX associate decision When the audit is submitted Then a defect should be automatically created and notification e-mail should be triggered to LoadX associate, Audit associate, Ops Manager and Audit Manager	**Scenario 2** Given a defect is triggered When notification email is sent Then the email should include - Job ID, Job Link, Associate ID, Associate decision, Audit Associate ID, Audit Decision and Comment	**Scenario 3** Given a defect is triggered When notification email fails to send Then system should log the failure and allow retry and tracking of unsent notifications

Operations and Audit Manager User Stories

US12	As an Ops Manager, I want jobs to be automatically assigned to associates, so that manual allocation is eliminated		
	Acceptance Criteria		
	Scenario 1 Given associates are logged in and active When jobs are available in the queue Then the jobs should be automatically assigned to available associates	**Scenario 2** Given no associates are logged in When jobs are available Then assignment should not be attempted and jobs should remain in the queue	**Scenario 3** Given multiple associates are logged in and active When jobs are assigned Then jobs should be distributed across associates to balance the workload

US13	**As an Ops manager, I want to view real-time KPIs across team, associates, geography and time, so that I can monitor team performance and take timely corrective actions**		
	Acceptance Criteria		
	Scenario 1 Given Ops Manager is logged into the tool When they access dashboard Then key KPIs (Accuracy, TAT, Jobs processed, Expired Jobs, Productive Hours, Defect Category) should be displayed and data reflected should be of latest information	**Scenario 2** Given dashboard is displayed When the Ops Manager applies filters (team, associate, geography, time range) Then the KPI data should also update based on filters	**Scenario 3** Given no data exits for the selected filters When the Ops Manager view dashboard Then the system should display a “No Data Available” message
US14	**As an Ops Manager, I want all job, audit and performance data to be stored in a centralized system, so that I can access accurate and consistent data for reporting and analysis**		
	Acceptance Criteria		
	Scenario 1 Given jobs are processed by associates and audits are completed When the data is generated Then all job, audit and performance data should be stored in a centralized system	**Scenario 2** Given data is stored in a centralized system When Ops Manager access reports or dashboards Then data should be available and reflect real - time data & have an option to export the data in excel & pdf	
US15	**As an Audit Manager, I want to configure audit sampling percentage, so that I can control level of quality checks**		
	Acceptance Criteria		
	Scenario 1 Given Audit Manager logged into tool When they enter a valid sampling percentage (between 10-15%) and save the configuration Then system should store and update sampling percentage and apply it to future job assignments	**Scenario 2** Given the Audit Manager is configuring the sampling percentage When they enter an invalid sampling percentage and save the configuration Then the system should display a validation error and prevent saving	**Scenario 3** Given the Audit Manager updates the sampling value When they do not save the changes Then system should retain the previously configured percentage

US16	As an Audit Manager, I want to monitor accuracy and defect trends, so that I can identify issues and improve overall process quality		
	Acceptance Criteria		
	Scenario 1 Given Audit manager is logged into tool When they access the dashboard Then accuracy percentage should be displayed and defect trends should be visible over a period of time	**Scenario 2** Given the dashboard is displayed When the Audit Manager applies filter (date range, location, team, associate) Then the accuracy and defect data should update based on selected filters	**Scenario 3** Given no data is available for selected filters When the Audit Manager views the dashboard Then the system should display a “No data available” message

Appendix- B

UAT Test Cases

TC - ID	US - ID (Reference)	High Level Scenario	Expected Outcome	Tester Role	Priority
TC-01	US-01	Associate opens an assigned job & views the screen	Image loads, with inline decision options	LoadX Associate	Critical
TC-02	US-02	Associate selects a single decision & submits job	Selected decision is recorded successfully	LoadX Associate	Critical
TC-03	US-03	Associate submits a job & continues workflow	Next job is automatically assigned & displayed	LoadX Associate	Critical
TC-04	US-04	Associate flags an image as poor quality	Recorded image quality and removed from queue	LoadX Associate	Critical
TC-05	US-05	Associate process a job while timer is active and within 2 mins submits the job	Timer starts as soon as job loads and submission is recorded	LoadX Associate	Critical
TC-06	US-06	Associate tries to process job after 2 mins	Job expires, removed from queue and reassigned to another associate	LoadX Associate	Critical
TC-07	US-07	Associate starts and ends a session	Session start and end time are recorded	LoadX Associate	Critical
TC-08	US-09	Audit associate start a session to process jobs	Screen displays image, along with job, associate_id and associate decision details	Audit Associate	Critical
TC-09	US-10	Audit associate submit audit decision for a job where there is no decision mismatch	Audit decision is recorded successfully	Audit Associate	Critical
TC-10	US-12	System assigns jobs to multiple active associates	Jobs are automatically distributed among loggedin associate	Ops Manager	Critical
TC-11	US-13	Ops Manager views dashboard for performance monitoring	Display of real-time KPIs	Ops Manager	Critical
TC-12	US-14	System stores and retrieves operational data	All job, audit and performance data are stored centrally and can be accessed	Ops Manager	Critical
TC-13	US-15	Audit Manager configures audit sampling percentage with valid percentage	Percentage is saved and applied to future jobs	Audit Manager	High

Appendix- C

Data Dictionary

Data Point	Description/Definition	Source/Generated By
Image ID/URL	Reference to truck load image	Data Pipeline
Upload_Timestamp	Time when the image was uploaded	System
Job ID	Unique identifier for each job/image	System (auto generated)
Assigned_Associate_ID	ID of associate assigned to job	System (auto assignment)
Job_Assigned_Timestamp	Time when job is assigned	System
Job_Status	Current state - Assigned/Completed/Expired/Flagged	System
Decision	Decision options - Complete Load Secured, Partial Load secured - Loose straps, few safety tags, partially covered with wrapping cover, Not secured - no straps, no safety tags, no wrapping cover, Overloaded Truck, Obstructed view	LoadX associate
Job_Submitted_Timestamp	Time of decision submission	System
Is_Flagged	Indicates if image is poor quality	LoadX associate
Expiry_Timestamp	Time when SLA breach occurs	System
Is_Expired	Indicates if job exceeded SLA	System
Reassignment_Count	Number of times a job is reassigned	System
Session_ID	Unique Session Identifier	System
Session_Start_Time	Time when session started	System
Session_End_Time	Time when session ended	System
Session Duration	Total duration between a session start and session end	System (calculated)
Jobs_Processed	Total jobs handled in a session. Includes flagged & expired	System (calculated)
Audit_ID	Unique ID for an audit job	System
Audit_Associate_ID	ID of audit associate	System
Audit_Decision	Decision taken by audit associate	Audit Associate
Audit_Timestamp	Time of audit completion	System
Is_mismatch	Indicates mismatch between decisions	System
Audit_Comment	Mandatory comment for mismatch	Audit associate
Defect_ID	Unique defect identifiers generated when there is audit decision mismatch	System
Defect_Created_Time	Time when defect is created	System
Defect_Category	Category under which a defect falls into - Wrong decision, Image misinterpretation, Incorrect Flagging,	System
Sampling_Percentage	% of jobs sent for audit	Audit Manager
Config_Effective_From	Start date of sampling configuration	System
Associate_Role	Role (LoadX/Audit/Manager)	System
Associate_ID	Unique ID of associate	System
Team	Identified by managers. Manager leads a team of associates	System
Associate_Location	Location of associate	System
Job_Location	Hub from which image is generated	System
Shift	Work shift during which session is active (Morning/Afternoon/Evening/Night)	System
Ops_Manager_ID	Unique ID of ops manager	System
Audit_Manager_ID	Unique ID of audit manager	System

Jobs_assigned_count	Total jobs assigned to associate	System (calculated)
Jobs_completed_count	Total jobs completed. Excludes flagged & expired	System (calculated)
TAT	Time taken for associate from job assignment to submission	System (calculated)
Accuracy	Percentage of correct decision jobs out of completed jobs	System (calculated)
SLA_Breach_Flag	Indicates SLA violation, yes/no	System
Defect count	Number of defect jobs out of completed jobs	Derived
Productivity Hours	Number of hours associates spent on tool processing jobs	Derived
Productivity rate	Number of jobs completed per unit time	System(calculated)